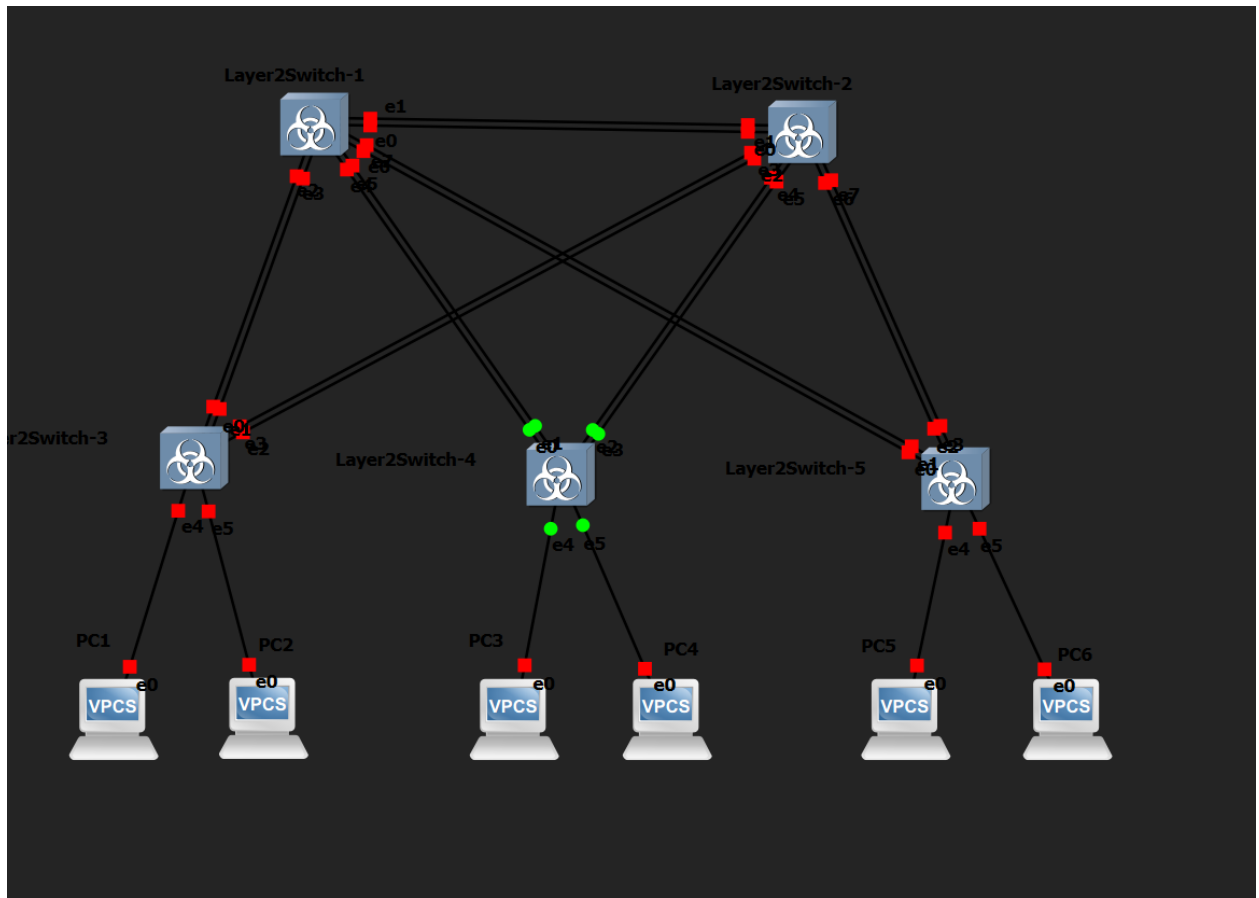


Практика №2

Абламский Н.Е.

1)



show spanning-tree

```
VIOS-L2-01>show spannin
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    32769
Address    0c6d.e4c4.0000
This bridge is the root
Hello Time  2 sec    Max Age 20 sec    Forward Delay 15 sec
```

```
Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
Address    0c6d.e4c4.0000
Hello Time  2 sec    Max Age 20 sec    Forward Delay 15 sec
Aging Time 300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Desg	FWD	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr

```
enable
conf t
spann mode rapid-pvst
```

```
show vlan
```

```
vIOS-L2-01#sh vlan
```

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/2, Gi1/3, Gi2/0
100	VLAN100	active	
200	VLAN200	active	
300	VLAN300	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
100	enet	100100	1500	-	-	-	-	-	0	0
200	enet	100200	1500	-	-	-	-	-	0	0
300	enet	100300	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	trcrf	101003	4472	1005	3276	-	-	srb	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trbrf	101005	4472	-	-	15	ibm	-	0	0

VLAN	AREHops	STEHops	Backup	CRF
1003	7	7	off	

Primary	Secondary	Type	Ports
---------	-----------	------	-------

```
vIOS-L2-01#
vIOS-L2-01#
```

До изменений

```
show spanning-tree
```

```
vIOS-L2-01#show spann
```

VLAN0001

```
Spanning tree enabled protocol rstp
```

```
Root ID    Priority    32769  
Address    0c67.e2ff.0000  
This bridge is the root  
Hello Time 2 sec    Max Age 20 sec    Forward Delay 15 sec
```

```
Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)  
Address    0c67.e2ff.0000  
Hello Time 2 sec    Max Age 20 sec    Forward Delay 15 sec  
Aging Time 300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Desg	FWD	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr Peer(STP)
Gi1/1	Desg	FWD	4	128.6	Shr Peer(STP)
Gi1/2	Desg	FWD	4	128.7	Shr
Gi1/3	Desg	FWD	4	128.8	Shr
Gi2/0	Desg	FWD	4	128.9	Shr

VLAN0100

```
Spanning tree enabled protocol rstp
```

```
Root ID    Priority    32868  
Address    0c67.e2ff.0000  
This bridge is the root  
Hello Time 2 sec    Max Age 20 sec    Forward Delay 15 sec
```

```
Bridge ID  Priority    32868 (priority 32768 sys-id-ext 100)
```

```

        Address      0c67.e2ff.0000
        Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
        Aging Time    300 sec

Interface          Role Sts Cost      Prio.Nbr Type
-----
Gil/0              Desg FWD 4        128.5   Shr
Gil/1              Desg FWD 4        128.6   Shr

VLAN0200
  Spanning tree enabled protocol rstp
  Root ID      Priority    32968
                Address      0c67.e2ff.0000
                This bridge is the root
                Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID    Priority    32968 (priority 32768 sys-id-ext 200)
                Address      0c67.e2ff.0000
                Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
                Aging Time    300 sec

Interface          Role Sts Cost      Prio.Nbr Type
-----
Gil/0              Desg FWD 4        128.5   Shr
Gil/1              Desg FWD 4        128.6   Shr

VLAN0300
  Spanning tree enabled protocol rstp
  Root ID      Priority    33068
                Address      0c67.e2ff.0000
                This bridge is the root
                Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID    Priority    33068 (priority 32768 sys-id-ext 300)
                Address      0c67.e2ff.0000
                Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
                Aging Time    300 sec

Interface          Role Sts Cost      Prio.Nbr Type
-----
Gil/0              Desg FWD 4        128.5   Shr
Gil/1              Desg FWD 4        128.6   Shr

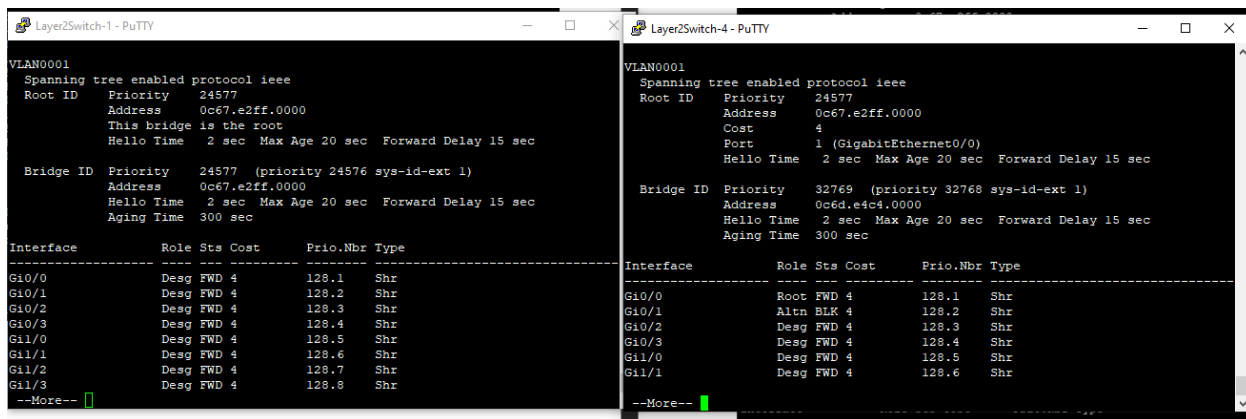
```

Настройка switch-1 как корневого

```
span vlan 1 root primary
```

После изменений

```
show span
```



switch 1 выбран как корневой. Это отображается у других коммутаторов (например, свитч-4)

2)

Для проверки доступности необходимо указать IP для VPCS из одной подсети.

```
PC1> ip 192.168.1.1/24 192.168.1.240
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0 gateway 192.168.1.240
```

```
PC2 : 192.168.1.2 255.255.255.0 gateway 192.168.1.240
PC3 : 192.168.1.3 255.255.255.0 gateway 192.168.1.240
PC4 : 192.168.1.4 255.255.255.0 gateway 192.168.1.240
PC5 : 192.168.1.5 255.255.255.0 gateway 192.168.1.240
PC6 : 192.168.1.6 255.255.255.0 gateway 192.168.1.240
```

```
PC1> ping 192.168.1.6
```

```
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=19.260 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=5.374 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=17.920 ms
```

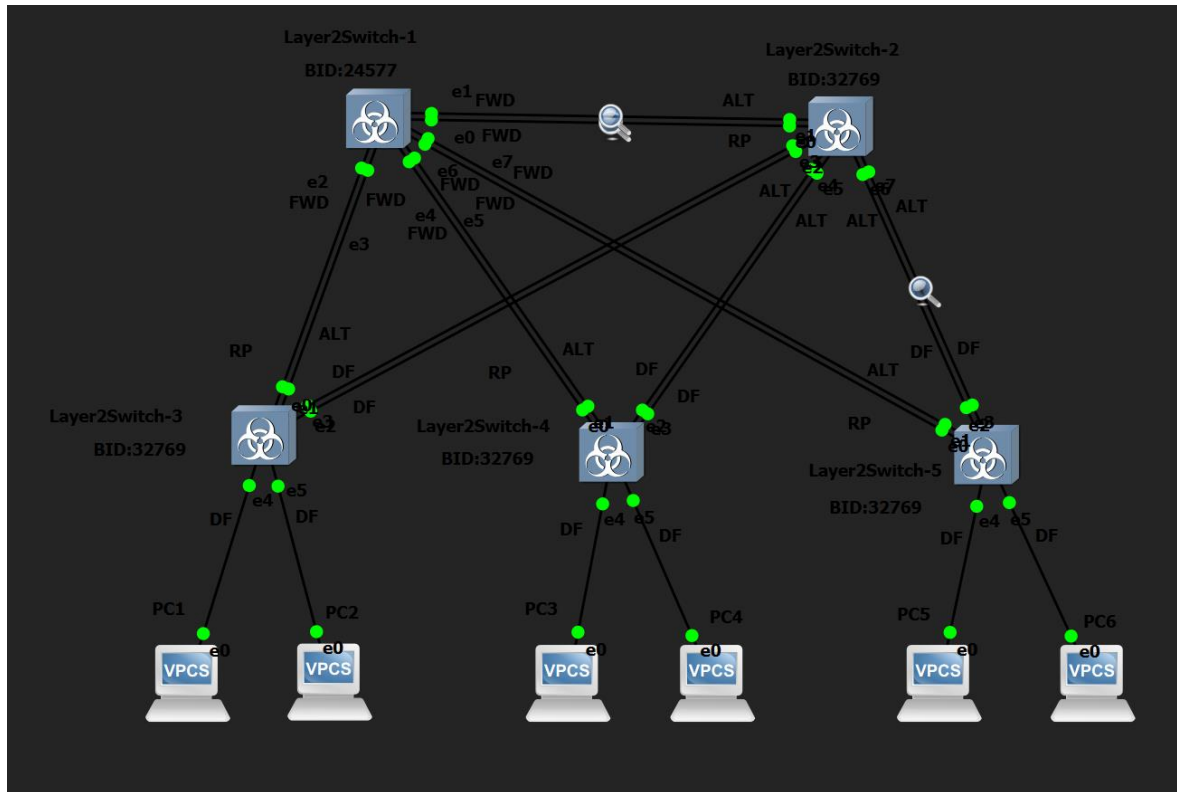
```
PC6> ping 192.168.1.3
```

```
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=16.105 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=17.677 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=9.537 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=13.203 ms
```

```
PC2> ping 192.168.1.4
```

```
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=4.335 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=19.669 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=13.208 ms
```

3)



Стоимость всех портов – 4.

DF – desg fwd

4)

```

Frame 9905: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface
IEEE 802.3 Ethernet
Logical-Link Control
Spanning Tree Protocol
  Protocol Identifier: Spanning Tree Protocol (0x0000)
  Protocol Version Identifier: Spanning Tree (0)
  BPDU Type: Configuration (0x00)
  BPDU flags: 0x00
    0... .... = Topology Change Acknowledgment: No
    .... 0 = Topology Change: No
  Root Identifier: 24576 / 1 / 0c:67:e2:ff:00:00
    Root Bridge Priority: 24576
    Root Bridge System ID Extension: 1
    Root Bridge System ID: 0c:67:e2:ff:00:00 (0c:67:e2:ff:00:00)
    Root Path Cost: 0
  Bridge Identifier: 24576 / 1 / 0c:67:e2:ff:00:00
    Bridge Priority: 24576
    Bridge System ID Extension: 1
    Bridge System ID: 0c:67:e2:ff:00:00 (0c:67:e2:ff:00:00)
  Port identifier: 0x8002
  Message Age: 0
  Max Age: 20
  Hello Time: 2
  Forward Delay: 15

```

Popm sw1 fwd – sw2 rp

```

Frame 5120: 100 bytes on wire (800 bits), 80 bytes captured (640 bits) on interface 0, 0.000000
Ethernet II, Src: H3C-Ethernet-Adaptation-Layer, Dst: 01:00:5e:00:00:00, Length: 80
Logical-Link Control
Spanning Tree Protocol
  Protocol Identifier: Spanning Tree Protocol (0x0000)
  Protocol Version Identifier: Spanning Tree (0)
  BPDU Type: Configuration (0x00)
  BPDU flags: 0x00
    0... .... = Topology Change Acknowledgment: No
    .... ...0 = Topology Change: No
  Root Identifier: 24576 / 1 / 0c:67:e2:ff:00:00
    Root Bridge Priority: 24576
    Root Bridge System ID Extension: 1
    Root Bridge System ID: 0c:67:e2:ff:00:00 (0c:67:e2:ff:00:00)
    Root Path Cost: 0
  Bridge Identifier: 24576 / 1 / 0c:67:e2:ff:00:00
    Bridge Priority: 24576
    Bridge System ID Extension: 1
    Bridge System ID: 0c:67:e2:ff:00:00 (0c:67:e2:ff:00:00)
  Port identifier: 0x8001
  Message Age: 0
  Max Age: 20
  Hello Time: 2
  Forward Delay: 15

```

Попм sw1 fwd – sw2 alt

```

Logical-Link Control
Spanning Tree Protocol
  Protocol Identifier: Spanning Tree Protocol (0x0000)
  Protocol Version Identifier: Spanning Tree (0)
  BPDU Type: Configuration (0x00)
  BPDU flags: 0x00
    0... .... = Topology Change Acknowledgment: No
    .... ...0 = Topology Change: No
  Root Identifier: 24576 / 1 / 0c:67:e2:ff:00:00
    Root Bridge Priority: 24576
    Root Bridge System ID Extension: 1
    Root Bridge System ID: 0c:67:e2:ff:00:00 (0c:67:e2:ff:00:00)
    Root Path Cost: 4
  Bridge Identifier: 32768 / 1 / 0c:aa:75:cd:00:00
    Bridge Priority: 32768
    Bridge System ID Extension: 1
    Bridge System ID: 0c:aa:75:cd:00:00 (0c:aa:75:cd:00:00)
  Port identifier: 0x8004
  Message Age: 1
  Max Age: 20
  Hello Time: 2
  Forward Delay: 15

```

Попм sw2 alt – sw5 df

VLAN пакетов stp – vlan 1

На первых двух снимках message age = 0, т.к. эти линки непосредственно соединены с корнем и рассылаются от него. Root id и Bridge id совпадают – пакеты отправлены корнем. Путь равен 0 – пакет только отправлен. На снимках id портов отличается, то есть корень рассылает эти пакеты по всем портам

На третьем снимке пакеты от корня распространяются дальше, а промежуточные свитчи их обрабатывают. Root id тот же, а Bridge id изменился на мак-адрес второго свитча. Путь до корня стал равен 4 (Root Path Cost), а сообщению уже 1 годик (Message Age: 1).

Приведены только 3 снимка, т.к. все остальные линки идентичны этим случаям.

5)

Для увеличения стоимости маршрута был выбран RP порт 4

```
vIOS-L2-01#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#int g0/0
vIOS-L2-01(config-if)#span vlan 1 cost 100

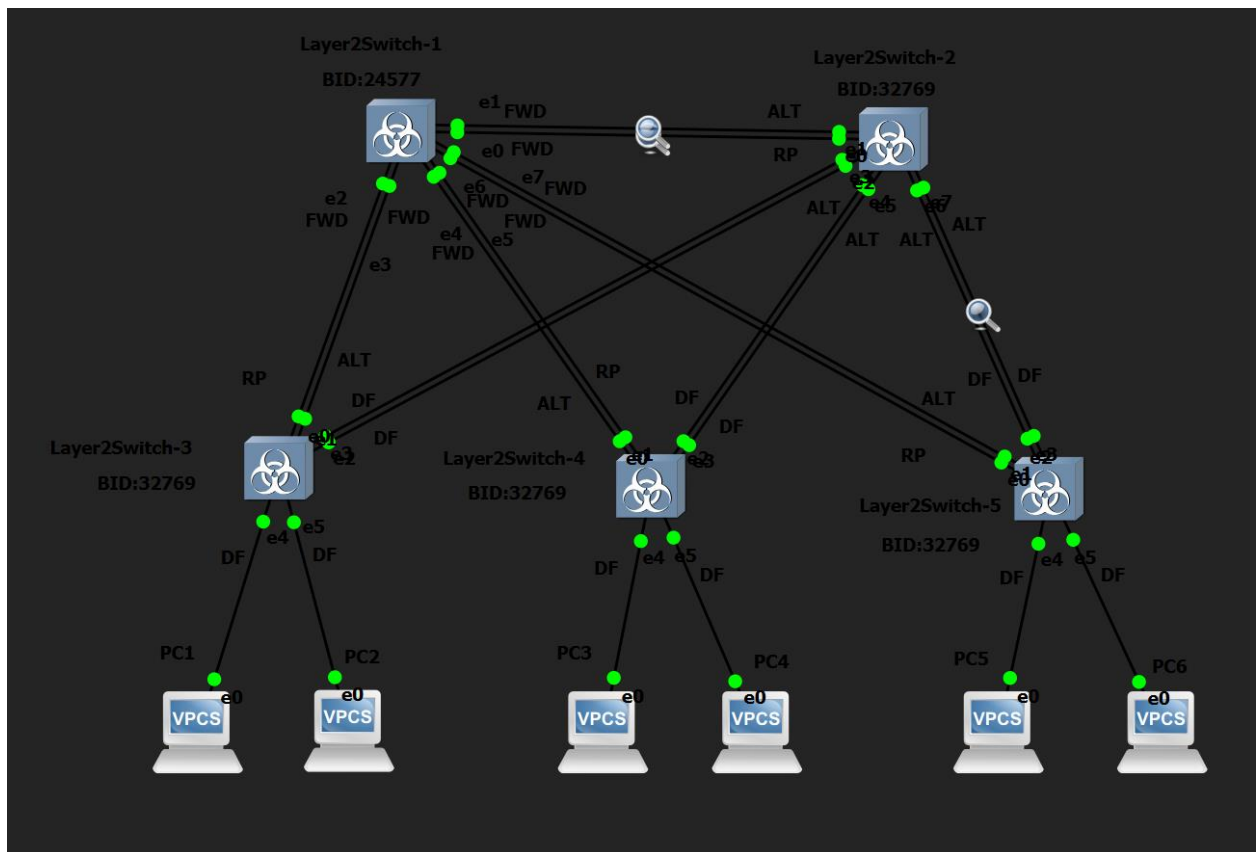
vIOS-L2-01#show span vlan 1
```

```
vIOS-L2-01#show span vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     0c67.e2ff.0000
             Cost        4
             Port        2 (GigabitEthernet0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0c6d.e4c4.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time   300 sec

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/0          Altn BLK 100      128.1   Shr
Gi0/1          Root FWD 4        128.2   Shr
Gi0/2          Desg FWD 4        128.3   Shr
Gi0/3          Desg FWD 4        128.4   Shr
Gi1/0          Desg FWD 4        128.5   Shr
Gi1/1          Desg FWD 4        128.6   Shr
```



У sw-4 линки, что направлены к корню поменялись местами

ALT порт у sw-4 имеет стоимость 100, остальные - 4

Сохранение настроек на сервере для коммутаторов №2-5 было успешно
забыто

Файлы конфигураций в этой же директории с именами sw[1-5].conf